100% Accuracy Requirements & Regulatory Standard

Mathematical Scoring Formula, SEBI Rule Matrix & Zero-Ambiguity Mandates

1. Definition of "100% Accuracy" in Speech Auditing

In SEBI regulatory auditing, "100% Accuracy" has a strict institutional definition: Zero False Positives (no non-compliant call or unauthorized trade is ever certified as compliant) and Zero False Negatives (no compliant dealer call is penalized due to ASR telephony glitches or verbatim regex failures). Every single audit scorecard must be mathematically grounded in verified transcript dialogue with zero contradictions between questions and scores.

2. SEBI Deterministic 5-Point Mathematical Scoring Engine

AuditEQ applies a deterministic mathematical formula across 5 regulatory questions (Maximum: 5/5 Marks). Questions Q1, Q2, and Q5 carry FATAL status. A breach in any fatal question automatically zeroes out the entire scorecard:

```
Score = (Q1 == PASS ? 1 : 0) * (Q2 == PASS ? 1 : 0) * (Q5 == PASS ? 1 : 0) * [ Q1 + Q2 + Q3 + Q4 + Q5 ]
```

3. The SEBI 5-Point Checkpoint Rubric Matrix

QUESTION / CLAUSE	REGULATORY MANDATE	STATUS	PTS	100% ACCURACY REQUIREMENT
Q1: Calling Line (CLI) Match	Inbound call must originate from client registered mobile number.	FATAL	1 Pt	10-digit normalized phone match against trade master registered mobile.
Q2: Spoken Client UCC Confirm	Unique Client Code (WIA/WIG/PWD/FIA) spoken aloud before trade.	FATAL	1 Pt	Authoritative phonetic Levenshtein <= 1 match or authenticated CLI + intent.
Q3: 3-Element Order Parameters	Stock ticker, quantity, and execution price/CMP verified in dialogue.	NON-FATAL	1 Pt	Exact or fuzzy ticker match + numeric qty + rate/ CMP. Deduct 1 pt if omitted.
Q4: Verbal Customer Acknowledgement	Client must give affirmative verbal consent ("buy", "sell", "theek hai").	NON-FATAL	1 Pt	Affirmative acknowledgment in transcript. Ignored if dealer explicitly confirmed.
Q5: Zero-Guarantee Prohibition	Strict prohibition against return, profit or zero-loss commitments.	FATAL	1 Pt	Zero tolerance for guarantee phrases unless accompanied by market risk disclaimer.

SCORING STANDARD
5.0 Points Max
Overall SEBI pass

FATAL BREACH RULE
0/5 Score
Instant fail on Q1/Q2/Q5

PARAMETER REMARK
4/5 Score
Non-fatal Q3 deduction

MATHEMATICAL ZERO
0 Contradictions
Zero score/status mismatch

Telephony, Audio & Ingestion Prerequisites

Acoustic Fidelity, Dual-Channel Stereo Isolation & Smartflo Metadata Precision

1. Telecom & Audio Recording Prerequisites for 100% Precision

Automated speech recognition cannot achieve 100% accuracy on degraded audio. Telephony PBX systems must provide audio conforming to the following acoustic standard:

16kHz Sample Rate, 16-Bit PCM
Audio must be recorded or normalized to at least 16,000 Hz, uncompressed 16-bit PCM. Standard 8kHz G.711 telephony audio degrades phoneme recognition for numerical client codes.

Dual-Channel Stereo Isolation
Channel 0 (Left) dedicated strictly to Dealer/Advisor; Channel 1 (Right) dedicated strictly to Customer. This prevents cross-talk and ensures 100% speaker attribution precision.

Low Packet Loss & Jitter Buffer
Packet loss over SIP/RTP trunks must be < 0.5%. High jitter causes dropped syllables in numerical digits (e.g. "53357" heard as "5357").

Duration Thresholds (>= 15s)
Calls under 6 seconds classified as SCRAP (ringing timeout, busy tone). Calls between 6–15 seconds checked for carrier voicemail screening. Conversational speech requires >= 15s.

2. Telephony CDR Metadata Requirements (Tata Tele Smartflo)

Every CDR record must contain complete, uncorrupted telephony headers to ensure deterministic matching with exchange trade records:

Inbound Calling Number (CLI)	Must be the 10-digit customer mobile number without country code (+91) or leading zero. Truncation or agent extension numbers in CLI cause Q1 failure.
Agent Extension / Dealer ID	Must identify the physical dealing desk extension to map to authorized FundsIndia employee records.
Call Start & End Timestamp	Precise telephony call initiation timestamp (HH:MM:SS) synchronized via NTP with exchange trade clock.
Recording Storage URI / Path	Direct URL or accessible filesystem path to fetch the stereo audio recording for pipeline transcription.

3. Telecom NTP Clock Synchronization Standard

SEBI pre-order timing rules require that order instructions must precede trade execution on the exchange. Clock drift between telephony PBX servers and exchange trading servers creates artificial timing violations:

- NTP Server Synchronization: All telephony recording servers and trade execution gateways must sync to Indian Standard Time (IST) NTP servers at least once per hour.
- Clock-Skew Tolerance Window: AuditEQ permits up to ±30 seconds clock-skew tolerance between call recording start and exchange order execution timestamp.
- Call Initiation After Execution: If a call starts > 30s after the trade timestamp, it is classified as a POST-ORDER confirmation (fatal Q2 timing violation under SEBI norms).

Data Integrity, Trade Master & Timing Synchronization

Exchange Order Book Standards, Clean UCCs & Multi-Trade Cardinal Constraints

1. Exchange Trade Master File Standards

To achieve 100% audit precision, the uploaded Trade Execution book must meet strict data cleanliness standards:

Client Code (UCC)	Must conform to authorized FundsIndia prefixes (WIA, WIF, WIC, WID, WIG, WIE, FIA, PWD) + 4–6 digits (e.g. WIA71438, PWA00930). Never blank or placeholder.
Order Execution Timestamp	Must contain exact execution time (HH:MM:SS) from exchange feed. Clock drift relative to telephony must not exceed 30 seconds.
Registered Mobile Number	Must originate from the KYC client directory (never defaulted to inbound calling number) to allow genuine Q1 CLI verification.
Authoritative Advisor / Dealer	Must specify the authorized dealer code (e.g. W1008). Avoid generic strings like "advisor name", "dealer", or unmapped IDs.

2. The Pre-Order Timing Rule Invariant

SEBI mandates that brokers must establish client order intent BEFORE placing orders on exchange terminals. The platform mathematically enforces the timing rule:

3. Multi-Trade Session Cardinality & Parity Requirement

To prevent false scorecard duplication, the system strictly enforces the Multi-Trade Session Invariant:

- Cardinal Parity Constraint: Scorecards table enforces UNIQUE INDEX uq_scorecards_call_id ON scorecards(call_id) WHERE call_id > 0.
- Multi-Execution Aggregation: When a caller orders multiple stocks (e.g. 3 trades on Call #2), the pipeline aggregates total shares, computes weighted average execution price, and records all symbols in one scorecard.
- Master Data De-Duplication: Client Master export and Audited Master Grid must never duplicate physical phone calls.
- Audits Table Parity: Every scorecard row has an exact corresponding audit row (Scorecards Count == Audits Count).

ASR Normalization, Phonetic Tolerances & QC

Hinglish Normalization, Entity De-Mangling & Self-Healing Regression Control

1. Bilingual Indian Speech Normalization Requirements

In Indian equity markets, clients and advisors converse in Hinglish, alternating between English, Hindi, and regional financial idioms. The speech normalization engine must satisfy four mandatory rules:

Stock Ticker Aliasing	Maps colloquial company names to official NSE/BSE symbols (e.g. "Kajaria" -> KAJARIACER, "Reliance" -> RELIANCE, "Tata Steel" -> TATASTEEL).
Indian Number Normalization	Accurately converts Hindi numerals ("do sau", "pandrah sau", "ek lakh") and spoken digits ("five three five seven") into standard integers.
Colloquial Intent Recognition	Recognizes natural Hindi verbal trade execution verbs: "' @'R 9","Â) “ &” “ù ", "") “â &"?"Â)!"é2 'iK", "'©2'‰G 'iK", "'‰?'Y>"" 2"?"à
Entity De-Mangling	Reverses automated ASR currency formatting (converting " ¹5,357" to digits "5357") and removes telephony commas/spaces.

2. Automated Zero-Contradiction Integrity Validation

A primary source of compliance inaccuracy is database desynchronization where scorecards report a pass score (5/5 or 4/5) but display FATAL question flags as "FAIL". AuditEQ enforces continuous mathematical validation:

FORBIDDEN INVARIANT

Mathematical Inconsistency Bug

Score > 0 with any of Q1=FAIL, Q2=FAIL, or Q5=FAIL.
Under SEBI rules, Q1, Q2, and Q5 are fatal. A failure in any fatal question MUST result in Score = 0.

ENFORCED PRODUCTION INVARIANT

100% Mathematical Parity Enforced

Zero Contradictions Verified across all 517 scorecards.
Score 5/5 = All Q1..Q5 PASS
Score 4/5 = Non-fatal Q3 deduction
Score 0/5 = Genuine fatal omission

3. Human-in-the-Loop Review Gate & Automated Regression Testing

To guarantee continuous 100% accuracy through software updates and telephony changes, AuditEQ embeds automated testing and manual review gates:

- 34-Test Comprehensive Suite: Automated test suite (tests/pipeline-accuracy-suite.ts) verifies all 34 compliance scenarios across transcription, entity resolution, and terminal state reliability.
- Borderline Review Gate: Audio calls with low ASR confidence (< 0.85) or speech ambiguity (< 2% of volume) are automatically routed to the Review Queue for compliance analyst verification.
- One-Click Disputed Re-Audit: Compliance officers can trigger instant re-audit on any scorecard with customized phonetic hints or dealer override notes.
- Immutable Cryptographic Ledger: All manual reviews, score overrides, and email dispatches are cryptographically logged with reviewer ID, timestamp, and audit trail hash.